Phan Ngoc Hoa

Software Engineer Intern

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Education

- **Ho Chi Minh City University of Technology (HCMUT) - VNUHCM** — Ho Chi Minh City, Viet Nam
  Pursuing a Bachelor of Computer Science — *October 2022 – Present*
  - GPA: 3.2

Skills

- **Programming languages:** C/C++, JavaScript, Python, Java
- **Front-end:** HTML, CSS, Tailwind CSS, React.js
- **Database Management Systems:** MySQL, MSSQL, Posgresql
- **Other Tools:** Git, Github, Figma, Postman, Latex, Docker
- **Cloud/Other:** Swagger, Kafka, gRPC

Project

- **Decentralized Federated Adaptive Weighting in Financial Systems** — Graduation Project
  Aggregation Researcher — *August 2025 – June 2026*
  - **Description:** Developed an innovative, privacy-preserving AI algorithm (DFedADP) optimized for detecting financial fraud across distributed networks with highly unbalanced . Additionally, engineered a full-stack banking application to simulate real-world transaction workflows.
  - **Technologies:** Python, Java Spring Boot, Next.js, P2PFL.
  - **Team Size:** 4
  - **Responsibilities:**
    - **Intelligent Weighting Mechanism:** Designed a smart scoring system that evaluates the reliability of AI updates from different distributed nodes, minimizing the impact of misleading or noisy data to ensure high model accuracy.
    - **Banking Simulation & System Integration:** Developed a full-stack banking simulation application adhering to the standard operational workflows of HSBC and OCB using Java Spring Boot and Next.js. This provided a functional environment to demonstrate real-time interactions, transaction pipelines, and data exchange with the decentralized fraud detection system.
    - **Comprehensive Performance Validation:** When evaluated across diverse data scenarios, the algorithm effectively reduced client drift on the MNIST dataset, achieving 95.71% accuracy under IID conditions and maintaining a robust 80.55% under extreme Non-IID settings ($\kappa = 0.1$)—outperforming standard decentralized methods by 39% to 69%. Furthermore, validation on a highly imbalanced simulated card transaction dataset demonstrated that the framework successfully balanced classification trade-offs, securing an optimal F1-Score of $\approx$ 0.38 and a critical Recall of $\approx$ 0.6.

Experiences

- **Blood Test Management System** — FPT Software Ho Chi Minh
  Back-End Developer — *May – Jyly 2025*
  - **Description:** Designed and implemented a microservices-based backend for blood sample processing. The system features three autonomous services handling Patient Management, User Management and Result Reporting, utilizing REST APIs for inter-service communication and distinct databases to ensure data isolation.
  - **Technologies:** Java Spring Boot.
  - **Responsibilities:**
    - **Design & Agile Collaboration:** Designed system workflows using Use Case diagrams and ERDs within an Agile Scrum framework, participating in weekly sprints to ensure timely delivery.
    - **API & Inter-service Communication:** Developed RESTful APIs with Swagger/OpenAPI documentation and engineered high-performance gRPC endpoints for efficient internal data retrieval.
    - **Asynchronous Processing:** Implemented a Message Broker to facilitate asynchronous data transfer, ensuring system decoupling and fault tolerance across microservices.

Honors and Awards

- **Second Place - Provincial Physics Contest, Dong Nai**